



Alzheimer Disease

(Alzheimer's Disease)

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Alzheimer disease is a progressive loss of mental function, characterized by degeneration of brain tissue, including loss of nerve cells, the accumulation of an abnormal protein called beta-amyloid, and the development of neurofibrillary tangles.

- Forgetting recent events is an early sign, followed by increasing confusion, impairment of other mental functions, and problems using and understanding language and doing daily tasks.
- Symptoms progress so that people cannot function, causing them to become totally dependent on others.
- Doctors base the diagnosis on symptoms and results of a physical examination, mental status tests, blood tests, and imaging tests.
- Treatment involves strategies to prolong functioning as long as possible and may include medications that may slow the progression of the disease.
- How long people live cannot be predicted, but death occurs, on average, about 7 years after the diagnosis is made.

(See also [Overview of Delirium and Dementia](#) and [Dementia](#).)

Alzheimer disease is a type of [dementia](#), which is a slow, progressive decline in mental function including memory, thinking, judgment, and the ability to learn.

In 60 to 80% of older adults with dementia, Alzheimer disease is the cause. It is rare among people younger than 65 years. It becomes more common with increasing age. In the United States, an estimated 10% of people aged 65 and over have Alzheimer disease. The percentage of people with Alzheimer disease increases with age:

- Age 65 to 74: 5%
- Age 75 to 84: 13%
- Age 85 and over: 33%

Alzheimer disease affects more women than men, partly because women live longer. The number of people with Alzheimer disease is expected to greatly increase as the proportion of older adults increases.

Causes of Alzheimer Disease

What causes Alzheimer disease is unknown, but genetic factors play a role: About 5 to 15% of cases run in families. Several specific gene abnormalities may be involved. Some of these abnormalities can be inherited when only one parent has the abnormal gene. That is, the abnormal gene is dominant. An affected parent has a 50% chance of passing on the abnormal gene to each child. About half of these children develop Alzheimer disease before age 65.

In most other cases, a single gene is not dominant. Instead, other genes affect the risk of getting Alzheimer disease. One gene abnormality affects apolipoprotein E (apo E)—the protein part of certain lipoproteins, which transport cholesterol through the bloodstream. There are 3 types of apo E:

- Epsilon-4: People with the epsilon-4 type develop Alzheimer disease more commonly and at an earlier age than other people.
- Epsilon-2: In contrast, people with the epsilon-2 type seem to be protected against Alzheimer disease.
- Epsilon-3: People with the epsilon-3 type are neither protected nor more likely to develop the disease.

However, genetic testing for apo E type cannot determine whether a specific person will develop Alzheimer disease. Therefore, this testing is not routinely recommended.

Risk factors, such as [high blood pressure](#), [diabetes](#), [high cholesterol levels](#), and [smoking](#), can increase the risk of Alzheimer disease. Treating these risk factors as early as midlife can reduce the risk of mental decline in older age.

Changes in the brain

In Alzheimer disease, parts of the brain degenerate, destroying nerve cells and reducing the responsiveness of the remaining ones to many of the chemical messengers that transmit signals between nerve cells in the brain (neurotransmitters). The level of acetylcholine, a neurotransmitter that helps with memory, learning, and concentration, is low.

Alzheimer disease causes the following abnormalities to develop in brain tissue:

- Beta-amyloid deposits: Accumulation of beta-amyloid (an abnormal, insoluble protein), which accumulates because cells cannot process and remove it
- Neuritic (senile) plaques: Clumps of dead nerve cells around a core of beta-amyloid
- Neurofibrillary tangles: Twisted strands of insoluble proteins in the nerve cell
- Increased levels of tau: An abnormal protein that is a component of neurofibrillary tangles and beta-amyloid

Such abnormalities develop to some degree in all people as they age but are much more numerous in people with Alzheimer disease. Doctors are not sure whether the abnormalities in brain tissue cause Alzheimer disease or result from some other problem that causes both the dementia and the abnormalities in brain tissue.

Researchers have also discovered that the abnormal proteins in Alzheimer disease (beta-amyloid and tau) resemble the abnormal proteins in [prion diseases](#). That is, they are misfolded and cause other proteins to misfold, causing the disease to progress.

Inflammation may also contribute to the development of Alzheimer disease. Inflammation has been observed in the brains of people with Alzheimer disease.

Did You Know...

- With aging, some brain abnormalities characteristic of Alzheimer disease develop in everyone.

Symptoms of Alzheimer Disease

Alzheimer disease causes many of the same symptoms as other dementias, such as the following:

- [Memory loss](#)
- Problems using language
- Changes in personality
- Disorientation
- Problems doing usual daily tasks
- Disruptive or inappropriate behavior

However, Alzheimer disease also differs from other dementias. For example, recent memory is typically affected much more than other mental functions.

Although when symptoms occur varies, categorizing them as early, intermediate, or late symptoms helps affected people, family members, and other caregivers have some idea of what to expect.

Personality changes and disruptive behavior ([behavior disorders](#)) may develop early or late in Alzheimer disease.

Early in Alzheimer disease

Symptoms develop gradually, so for a while, many people continue to enjoy much of what they enjoyed before developing Alzheimer disease.

Symptoms usually begin subtly. People whose disease develops while they are still employed may not do as well in their jobs. In people who are retired and not very active, the changes may not be as

noticeable.

The first and most noticeable symptom may be

- Forgetting recent events because forming new memories is difficult
- Sometimes changes in personality (people may become emotionally unresponsive, depressed, or unusually fearful or anxious)

Early in the disease, people become less able to use good judgment and think abstractly. Speech patterns may change slightly. People may use simpler words, a general word or many words rather than a specific word, or use words incorrectly. They may be unable to find the right word.

People with Alzheimer disease have difficulty interpreting visual and audio cues. Thus, they may become disoriented and confused. Such disorientation may make driving a car difficult. They may get lost on their way to the store. People may be able to function socially but may behave unusually. For example, they may forget the name of a recent visitor, and their emotions may change unpredictably and rapidly.

Many people with Alzheimer disease often have insomnia. They have trouble falling or staying asleep. Some people become confused about day and night.

At some point, psychotic behavior ([hallucinations](#), [delusions](#), or paranoia) develops in many people with Alzheimer disease.

Later in Alzheimer disease

As Alzheimer disease progresses, people have trouble remembering events in the past. They start to forget the names of friends and relatives. They may require help with eating, dressing, bathing, and going to the toilet. All sense of time and place is lost: People with Alzheimer disease may even get lost on their way to the bathroom at home. Their increasing confusion puts them at risk of wandering and falling.

Disruptive or inappropriate behavior, such as wandering, agitation, irritability, hostility, and physical aggression, is common.

Eventually, people with Alzheimer disease cannot walk or take care of their personal needs. They may be incontinent and unable to swallow, eat, or speak. These changes put them at risk of [undernutrition](#), [pneumonia](#), and [pressure sores](#) (bedsores). Memory is completely lost.

Ultimately, coma and death, often due to infections, result.

Behavior disorders in Alzheimer disease

Because people are less capable of controlling their behavior, they sometimes act inappropriately or disruptively (for example, by yelling, throwing, hitting, or wandering). These actions are called behavior disorders.

Several effects of Alzheimer disease contribute to this behavior:

- Because people with Alzheimer disease have forgotten the rules of proper behavior, they may act in socially inappropriate ways. When hot, they may undress in public. When they have sexual impulses, they may masturbate in public, use off-color or lewd language, or make sexual demands.
- Because they have difficulty understanding what they see and hear, they may misinterpret an offer of help as a threat and may lash out. For example, when someone tries to help them undress, they may interpret it as an attack and try to protect themselves, sometimes by hitting.
- Because their short-term memory is impaired, they cannot remember what they are told or have done. They repeat questions and conversations, demand constant attention, or ask for things (such as meals) they have already received. They may become agitated and upset when they do not get what they ask for.
- Because they cannot express their needs clearly or at all, they may yell when in pain or wander when lonely or frightened. They may wander, yell, or call out when they cannot sleep.

Whether a particular behavior is considered disruptive depends on many factors, including how tolerant the caregiver is and what sort of situation the person with Alzheimer disease is living in.

Progression of Alzheimer disease

Progression is unpredictable. People live, on average, about 7 years after the diagnosis is made. Most people with Alzheimer disease who can no longer walk live no more than 6 months. However, how long people live varies widely.

Diagnosis of Alzheimer Disease

- A doctor's evaluation
- Mental status testing
- Usually blood tests and imaging tests to rule out other causes

Diagnosis of Alzheimer disease is similar to that of other dementias.

Doctors must determine whether a person has dementia and, if so, whether the dementia is Alzheimer disease.

Doctors can usually diagnose Alzheimer disease based on the following:

- Symptoms, which are identified by asking the person and family members or other caregivers questions
- Results of a physical examination
- Results of a [mental status test](#)
- Results of additional tests, such as blood tests, computed tomography (CT), or magnetic resonance imaging (MRI)

Diagnosis of dementia

[Mental status testing](#), consisting of simple questions and tasks, helps doctors determine whether people have dementia.

Neuropsychologic testing, which is more detailed, is sometimes needed. This testing covers all the main areas of mental function, including mood, and usually takes 1 to 3 hours. This testing helps doctors distinguish dementia from other conditions that can cause similar symptoms, such as [age-associated memory impairment](#), [mild cognitive impairment](#), and [depression](#).

Information from the above sources can usually help doctors rule out delirium as the cause of symptoms (see table [Comparing Delirium and Dementia](#)). Doing so is essential because delirium, unlike dementia, can often be reversed if promptly treated. Differences between the disorders include the following:

- Dementia affects mainly memory, and delirium affects mainly attention.
- Dementia typically begins gradually and has no definite beginning point. Delirium begins suddenly and often has a definite beginning point.

Diagnosis of Alzheimer disease

Alzheimer disease is suspected when the following are present:

- The diagnosis of dementia has been confirmed.
- Usually, the most noticeable symptom, particularly in the beginning, is forgetting recent events or not being able to form new memories.
- Memory and other mental functions have gradually deteriorated and are continuing to deteriorate.
- The dementia began after age 40 and usually after age 65.
- Doctors have ruled out other brain disorders (such as a brain tumor or stroke) that could be causing the problems.

Some symptoms can help doctors distinguish Alzheimer disease from other dementias. For example, visual hallucinations (seeing things or people that are not there) are more common and occur earlier in [dementia with Lewy bodies](#) than in Alzheimer disease. Also, people with Alzheimer disease are often better-groomed and neater than people with other dementias.

Information from additional tests helps doctors make the diagnosis of Alzheimer disease and exclude other types and causes of dementia.

Analysis of cerebrospinal fluid (CSF), obtained during a [spinal tap](#), and [positron emission tomography](#) (PET) may be used to help diagnose Alzheimer disease. If CSF analysis detects a low level of beta-amyloid and if PET scans show amyloid deposits or tau deposits in the brain, the diagnosis is more likely to be Alzheimer disease. However, these tests are not routinely available.

The diagnosis of Alzheimer disease can be confirmed only when a sample of brain tissue is removed (after death, during an autopsy) and examined under a microscope. Then, the characteristic loss of

nerve cells, neurofibrillary tangles, and neuritic plaques containing beta-amyloid can be seen throughout the brain, particularly in the area of the temporal lobe that is involved in forming new memories.

Treatment of Alzheimer Disease

- Safety and supportive measures
- Medications that may improve mental function

Treatment of Alzheimer disease involves general measures to provide safety and support, as for all dementias. Also, certain medications can help for a while. The person with Alzheimer disease, family members, other caregivers, and the health care practitioners involved should discuss and decide on the best strategy for that person.

Pain and any other disorders or health problems (such as a urinary tract infection or constipation) are treated. Such treatment may help maintain function in people with dementia.

Safety and supportive measures

Creating a safe and supportive environment can be very helpful.

Generally, the environment should be bright, cheerful, safe, stable, and designed to help with orientation. Some stimulation, such as a radio or television, is helpful, but excessive stimulation should be avoided.

Structure and routine help people with Alzheimer disease stay oriented and give them a sense of security and stability. Any change in surroundings, routines, or caregivers should be explained to people clearly and simply.

Following a daily routine for tasks such as bathing, eating, and sleeping helps people with Alzheimer disease remember. Following a regular routine at bedtime may help them sleep better.

Activities scheduled on a regular basis can help people feel independent and needed by focusing their attention on pleasurable or useful tasks. Such activities should include physical and mental activities. Activities should be broken down in small parts or simplified as the dementia worsens. (See also [Creating a Beneficial Environment for People With Dementia.](#))

Medications

The **cholinesterase inhibitors** donepezil, galantamine, and rivastigmine increase the level of acetylcholine in the brain. (Acetylcholine is a neurotransmitter that helps with memory, learning, and concentration.) This level may be low in people with Alzheimer disease. These medications may temporarily improve mental function, including memory, but they do not slow the progression of the disease. Only some of the people who have Alzheimer disease benefit from these medications. For these people, the medications may effectively turn the clock back 6 to 9 months. These medications are

most effective in people with mild to moderate disease. The most common side effects include nausea, vomiting, weight loss, and abdominal pain or cramps.

Memantine appears to help relieve symptoms of moderate to severe Alzheimer disease. Memantine can be used with a cholinesterase inhibitor.

Antiamyloid monoclonal antibody therapies, including lecanemab and donanemab, may help slow progression of Alzheimer disease by removing amyloid plaque. These medications are given by injection into the bloodstream (intravenous infusion) every 2 to 4 weeks. In some people, the medications can cause some swelling and small hemorrhages in the brain, and people receiving these medications should have brain MRIs at intervals to monitor for these changes.

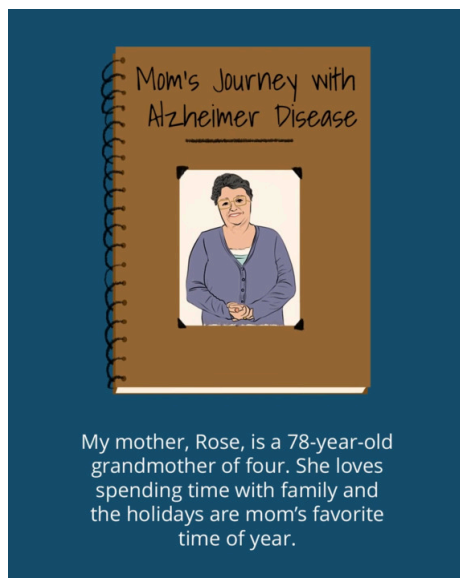
Researchers continue to study **other medications** that may prevent or slow the progression of Alzheimer disease. Estrogen therapy for women, nonsteroidal anti-inflammatory drugs (NSAIDs, such as ibuprofen or naproxen), and ginkgo biloba have been studied, but none has consistently proved to be effective. Moreover, estrogen appears to do more harm than good.

Vitamin E is an antioxidant that theoretically might help protect nerve cells from damage or help them function better. Whether vitamin E is useful is unclear.

Before people take any dietary supplement, they should discuss the risks and benefits with their doctor.

Hidden Disability: Alzheimer D...

VIDEO



Care for caregivers

Caring for people with Alzheimer disease is stressful and demanding, and caregivers may become depressed and exhausted, often neglecting their own mental and physical health (see also table [Caring for Caregivers](#)). The following measures can help caregivers:

- **Learning about how to effectively meet the needs of people with Alzheimer disease and what to expect from them:** Caregivers can get this information from nurses, social workers, organizations, and published and online materials.
- **Seeking help when it is needed:** Caregivers can talk to social workers (including those in the local community hospital) about appropriate sources of help, such as day-care programs, visits by home nurses, part-time or full-time housekeeping assistance, and live-in assistance. Counseling and support groups can also help.
- **Caring for self:** Caregivers need to remember to take care of themselves. They should not give up their friends, hobbies, and activities.

Long-term care

[Planning for the future](#) is essential because Alzheimer disease is progressive. Long before a person with Alzheimer disease needs to be moved to a more supportive and structured environment, family members should plan for this move and evaluate the options for long-term care. Such planning usually involves the efforts of a doctor, a social worker, nurses, and a lawyer, but most of the responsibility falls on family members.

Decisions about moving a person with Alzheimer disease to a more supportive environment involve balancing the desire to keep the person safe with the desire to maintain the person's sense of independence as long as possible.

Some long-term care facilities specialize in caring for people with Alzheimer disease. Staff members are trained to understand how people with Alzheimer disease think and act and how to respond to them. These facilities have routines that make the residents feel secure and provide appropriate activities that help them feel productive and involved in life. Most facilities have appropriate safety features. Finding a facility that has the appropriate safety features is important.

End-of-life issues

Before people with Alzheimer disease become too incapacitated, decisions should be made about medical care, and financial and legal arrangements should be made. These arrangements are called [advance directives](#). People should appoint a person who is legally authorized to make treatment decisions on their behalf (a health care proxy). They should discuss their [health care wishes](#) with this person and their doctor (see [Legal and Ethical Concerns](#)). Such issues are best discussed with all concerned long before decisions are necessary.

As Alzheimer disease worsens, treatment tends to be directed at maintaining the person's comfort rather than at attempting to prolong life.

Prevention of Alzheimer Disease

Some research tentatively suggests certain measures that may help prevent Alzheimer disease:

- **Keeping mentally active:** People are encouraged to continue doing activities that challenge the mind, such as learning new skills, doing crossword puzzles, and reading the news. These activities may promote the growth of new connections (synapses) between nerve cells and thus help delay dementia.
- **Exercising:** Exercising helps the heart function better and, for unclear reasons, may help the brain function better.
- **Controlling high blood pressure:** High blood pressure may damage blood vessels that carry blood to the brain and thus reduce the brain's oxygen supply, possibly disrupting connections between nerve cells.
- **Controlling blood sugar levels:** High blood sugar increases the risk of developing prediabetes (blood glucose levels that are too high to be considered normal but not high enough to be labeled diabetes) and [diabetes](#), which increase risk of Alzheimer disease.
- **Controlling cholesterol levels:** Some evidence suggests that having high cholesterol levels may be related to developing Alzheimer disease. Thus, people may benefit from a diet low in saturated fats and, if needed, medications (such as statins) to lower cholesterol and other fats (lipids).
- **Limiting alcohol intake to modest amounts:** Once dementia develops, abstaining from alcohol is usually best because it can make symptoms of dementia worse.

More Information

The following English-language resources may be useful. Please note that The Manual is not responsible for the content of these resources.

[Alzheimer's Association](#): Information about Alzheimer disease, including statistics, causes, risk factors, and symptoms. Also resources for support, including information about daily care of people with Alzheimer disease, care for the caregiver, and support groups.

[The Alzheimer's Society](#): A guide to dementia (including five things important things to know), a guide for caregivers, and information about types of dementia, symptoms, diagnosis, treatments, risk factors, and prevention.

[Institute of Neurological Disorders and Stroke Alzheimer's Disease Information Page](#): Information about treatment, prognosis, and available clinical trials and links to other organizations focused on Alzheimer disease.



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